Bug Report: Relationships of a Missing Project

Bug ID: PROJ-002

Summary: Relationship GET on a nonexistent project returns 200 with other projects' data.

Description:

GET and HEAD on `/projects/:id/tasks` and `/projects/:id/categories` do not check that the project exists. For an id that matches no project the API answers 200 and returns the tasks, or categories, linked to every project in the system. A todo that belongs to two projects appears twice.

Potential impact:

A client that asks for the tasks of a deleted or mistyped project receives a plausible list that belongs to other projects, with no error to signal the mistake. Decisions made from that list are based on wrong data.

Steps to reproduce:

1. Start the application with its default data, where project 1 has todos 1 and 2.
2. Send GET `/projects/0/tasks`.

Expected result: 404 with an error message, as GET `/projects/0` returns.

Actual result: 200 with todos 1 and 2 in the body.

Found in: Projects session; curl commands in `Exploratory testing scripts/Projects/projects_session.sh`, section "Behavior that differs from the documentation".

Unit tests: `getTasksOfMissingProjectReturnsNotFound`, `getCategoriesOfMissingProjectReturnsNotFound` (expected), `getTasksOfMissingProjectReturnsTheTasksOfOtherProjects`, `getCategoriesOfMissingProjectReturnsTheCategoriesOfOtherProjects` (actual). Expected tests are in `ProjectsDocumentedBehaviorTest`; actual tests are in `ProjectsActualBehaviorTest`.